Luhuan Wu

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Last updated: Jul 13, 2026

Academic Positions

Johns Hopkins University — Baltimore, MD
Department of Applied Mathematics and Statistics
Assistant Professor — 2026 - Current

Flatiron Institute — New York, NY
Center for Computational Mathematics
Associate Research Scientist — 2025 - 2026

Education

Columbia University — New York, NY
Ph.D. in Statistics. Advisors: John P. Cunningham and David M. Blei — 2020 - 2025
M.Phil. in Statistics — 2020 - 2024
M.S. in Data Science — 2018 - 2020

Nanjing University — Nanjing, China
B.S. in Mathematics — 2014 - 2018

Publications & Preprints

* for equal contribution

1. Robust Inference-Time Steering of Protein Diffusion Models via Embedding Optimization. *arXiv:2602.05285*
   Minhuan Li, Jiequn Han, Pilar Cossio, **Luhuan Wu**
2. Reverse Diffusion Sequential Monte Carlo Samplers. *NeurIPS 2025*
   Luhuan Wu, Han Yi, Christian A. Naesseth, John P. Cunningham
3. Bayesian Modeling of Data from Heteronegoues Environments. *arXiv:2506.22675*
   Luhuan Wu, Mingzhang Yin, Yixin Wang, John P. Cunningham, and David M. Blei
4. Posterior Uncertainty Quantification in Neural Networks using Data Augmentation. *AISTATS 2024*
   Luhuan Wu, Sinead Williamson
5. Practical and Asymptotically Exact Conditional Sampling in Diffusion Models. *NeurIPS 2023*
   Luhuan Wu*, Brian L. Trippe*, John P. Cunningham, and David M. Blei
6. Denoising Deep Generative Models. *NeurIPS 2022 I Can't Believe It's Not Better Workshop*
   Gabriel Loaiza-Ganem, Brendan Leigh Ross, **Luhuan Wu**, John P. Cunningham, Jesse C. Cresswell, Anthony L. Caterini
7. Variational Nearest Neighbor Gaussian Processes. *ICML 2022*
   Luhuan Wu, Geoff Pleiss, and John P. Cunningham
8. Bias-free Scalable Gaussian Processes via Randomized Truncations. *ICML 2021*
   Andres Potapczynski*, **Luhuan Wu***, Dan Biderman*, Geoff Pleiss, and John P. Cunningham

9. Hierarchical Inducing Point Gaussian Process for Inter-domian Observations. *AISTATS 2021*
Luhuan Wu*, Andrew Miller*, Lauren Anderson, Geoff Pleiss, David M. Blei, and John P. Cunningham
10. Inverse Articulated-body Dynamics from Video via Variational Sequential Monte Carlo. *NeurIPS 2020 Workshop on Differentiable Vision, Graphics, and Physics in Machine Learning*
Dan Biderman, Christian A. Naesseth, **Luhuan Wu**, Taiga Abe, Alice C. Mosberger, Leslie J. Sibener, Rui Costa, James Murray, John P. Cunningham
11. Variational Objectives for Markovian Dynamics with Backward Simulation. *ECAI 2020*
Antonio Khalil Moretti*, Zizhao Wang*, **Luhuan Wu***, Iddo Drori, Itsik Pe'er

Professional experience

Apple, AIML — Seattle, WA
Research Intern — May 2023 - Sept 2023

Columbia University, Zuckerman Institute — New York, NY
Research Staff — Feb 2020 - Sept 2020

Academic Service

Reviewer for JMLR (2021,2023,2025), ICML (2023,2024), NeurIPS (2022,2023,2025), ICLR (2024,2025,2026), AISTATS (2022,2023,2024), UAI (2024,2025), AAAI (2025), TMLR (2025), Statistics and Computing (2025, 2026), Electronic Journal of Statistics (2026)

Talks and Panel Discussions

Mar 2026: Monte Carlo Online Seminar. *Virtual*

Dec 2025: UK Probabilisitc AI "Prob_AI" Online Seminar. *Virtual*

Dec 2025: NeurIPS 2025 Workshop on Structured Probabilistic Inference & Generative Modeling. *San Diego, CA*

Nov 2025: Department of Statistics, University of Warwick. *Coventry, UK*

Oct 2025: Department of Statistics, Rutgers University. *New Brunswick, NJ*

Aug 2025: Microsoft Research New England Generative Modeling and Sampling Seminar. *Boston, MA*

Jun 2025: The Gatsby Tri-center Annaul Meeting. *London, UK*

Apr 2025: The 7th Advances in Approximate Bayesian Inference. *Singapore*

Jun 2024: The 33rd ICSA Applied Statistics Symposium. *Nashville, TN*

Apr 2023: Berkeley–Columbia Meeting in Engineering and Statistics. *New York, NY*

May 2022: The 35th New England Statistical Symposium. *Storrs, CT*